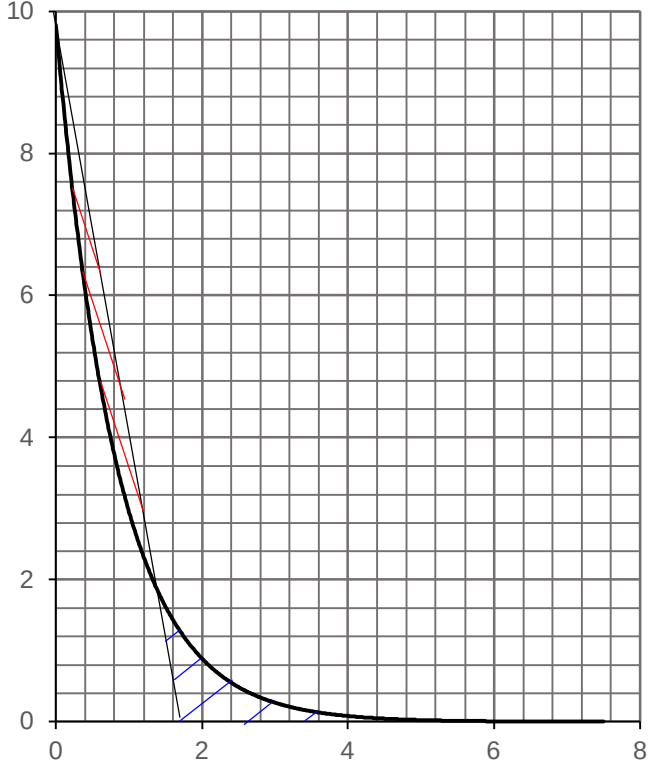
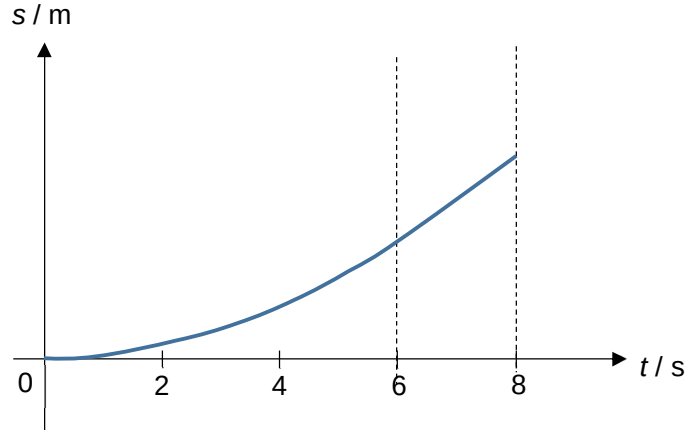


Qns	Answer	Marks
2(a)	Acceleration is the rate of change of velocity.	B1
(b)(i)	As the object accelerates downwards, its velocity increases with time. Since <u>the drag force (due to viscous fluid) on the object increases with speed,</u> <u>net force</u> (weight – resistive/drag force) acting on the object <u>decreases,</u> hence the acceleration decreases with time.	B1 B1
(b)(ii)	Initially the velocity of the object is zero, hence the <u>drag force is also zero.</u> The <u>acceleration of the object is only due to its weight / gravitational force ,</u> hence acceleration 9.8 m s^{-2} . (Assume upthrust is negligible since given that acceleration is 9.8 m s^{-2})	B1
(b)(iii)	Change in velocity = area under a-t graph $v - 0 \approx (50 \text{ small squares})(0.4 \times 0.4) = 8.0 \text{ m s}^{-1}$ (range: 7.5-9.0 m s^{-1}) One way to estimate the area under the curve is to draw a line as shown in the diagram below. The line chosen is such that the area shaded in red is roughly equal to the area shaded in blue. The area under the original curve is then roughly equal to the area of the triangle, with base 1.7 and height 9.8. Thus, velocity = area under curve $\approx \frac{1}{2} \times 9.8 \times 1.7 = 8.3 \text{ m s}^{-1}$. Better estimation of the area can be achieved by using more triangles/trapeziums to approximate it.  Some students used kinematics equations without understanding that the acceleration graph already shows that the acceleration is not constant.	C1 A1

(b)(iv)



Gradient = 0 at $t = 0$ s, then gradually increasing till $t \approx 5.2$ s .

After $t \approx 6$ s, gradient is constant (terminal velocity reached) at the value that was reached at 6 s - no kink allowed.

B1**B1****Qns****Answer****Marks**